

David Bean

DATA SCIENCE, DATA PIPELINES

Website: davidwbean.com | **LinkedIn:** linkedin.com/in/david98bean | **Email:** david98bean@gmail.com | **Technical Resume** 

SKILLS

Languages	Python, SQL, Bash, R
Acquisition	REST APIs, web scraping, JSON, requests, BeautifulSoup, web crawlers, CSV, geospatial data, NetCDF
Wrangling	pandas, NumPy, validation suites, merging sources, deduplication, SciPy, GeoPandas, dplyr, xarray
Pipelines	incremental processing, parallelization, larger-than-memory processing (100+ GB), checkpointing
Linux	parallel batch jobs, SSH, Slurm, grep, sed, awk, Make, tmux
Machine Learning	scikit-learn, XGBoost, feature selection, cross-validation, gradient boosting, PCA, k-means, UMAP
Statistics	hypothesis testing, linear regression, logistic regression, experimental design, t-tests, chi-square
Visualization	interactive dashboards, matplotlib, Google Sheets, Excel, Jupyter, ggplot2, Vega-Altair, RStudio
Databases	schema design, joins, DDL, ER diagrams, GROUP BY, primary and foreign keys
Version Control	Git, GitHub, pull requests, GitLab, branching and merging
Containers	Docker, Dockerfiles, GPU containers, Podman, Apptainer, venv

EDUCATION

Master of Science in Biomedical Informatics	May 2026
University of Utah Salt Lake City, UT Emphasis: Data Science and AI GPA: 3.95	
Bachelor of Science in Biology, magna cum laude	May 2023
Utah Tech University Saint George, UT Emphasis: Biomedical Sciences GPA: 3.80	

EXPERIENCE

Research Assistant II	Jun 2025 - Dec 2025
University of Utah - Department of Biomedical Informatics, Salt Lake City, UT	
• Diagnosed a published mutation classifier's precision drop from 0.71 to 0.02 on patient data as sensitivity to class ratio • Optimized pipeline runtime 10x (~50 to 5 hours) via parallel, containerized Slurm jobs and a resumable task scheduler • Eliminated ~90% of harmless mutations per patient via population databases, retaining 96% of known cancer mutations • Defended investment in the research to department leadership in a 30-minute talk distilling 6 months of work	
Research Assistant	Oct 2023 - Apr 2024
Utah Tech University - Klabacka Lab, Saint George, UT	
• Automated retrieval of protein data for 4,747 genes across 5 species by REST API in a Dockerized Bash and Python pipeline • Batch-generated data visualizations in R (ggplot2), looping one reusable chart design over data from 141 bacterial species • Demonstrated a vital protein ranked below 56% of 2,257 genes in genetic similarity, highlighting key positions for survival • Unified code from multiple contributors into one automated pipeline and presented findings at 3 research poster sessions	
Laboratory Instructor, Biology 1625	Aug 2023 - May 2024
Utah Tech University - Department of Biological Sciences, Saint George, UT	
• Led 3 lab sections of 18 to 24 students through full research cycles, from posing a question to presenting for peer feedback • Taught students to state null and alternative hypotheses, design experiments, and draw conclusions from statistical tests • Lectured on choosing the right statistical test for a dataset (paired and unpaired t-test, chi-square, linear regression) • Trained students to combine statistical tests with multiple visualizations to reveal patterns from more than one perspective	

PROJECTS

Interactive Monarch Migration Map 	2024
• Told the story of monarch migration in an interactive Vega-Altair dashboard of sightings and weekly temperature maps • Harmonized 227,000+ monarch sightings with the temperature grids by reprojecting both to the Mercator projection	
Motorcycle Crash Risk Analysis 	2025
• Backed all 14 group comparison charts with hard numbers, a statistical test and p value behind each (Cliff's delta 0.75) • Merged 7 years of federal crash files on a composite key into 870,000+ person-level records with pandas	